

JAVA PROGRAMMING LAB

Experiment No.: 56

Date: 13/08/2026

Aim:

WAP to accept and display employee details such as employee ID, name and salary.

Program:

```
1 package Program;
2
3 import java.io.*;
4
5 class Employee {
6
7     int empid;
8     String empname;
9     float salary;
10
11     void getval() throws IOException {
12
13         BufferedReader br = new BufferedReader(
14             new InputStreamReader(System.in));
15
16         System.out.print("Enter empid: ");
17         empid = Integer.parseInt(br.readLine());
18
19         System.out.print("Enter empname: ");
20         empname = br.readLine();
21
22         System.out.print("Enter salary: ");
23         salary = Float.parseFloat(br.readLine());
24     }
25
26     void showval() {
27
28         System.out.println("\nEmployee Details");
29         System.out.println("Employee ID: " + empid);
30         System.out.println("Employee Name: " + empname);
31         System.out.println("Employee Salary: " + salary);
32     }
33 }
34
35 class EmployeeDemo {
36
```

```
37     public static void main(String[] args) throws IOException {  
38  
39         Employee e = new Employee();  
40  
41         e.getval();  
42         e.showval();  
43     }  
44 }
```

Output:

```
Enter empid: 101  
Enter empname: Raghu  
Enter salary: 75000  
Employee Details  
Employee ID: 101  
Employee Name: Raghu  
Employee Salary: 75000.0
```

JAVA PROGRAMMING LAB

Experiment No.: 57

Date: 13/08/2026

Aim:

WAP to accept the number of rows and columns of a matrix, read its elements and display its transpose.

Program:

```
1 package Program;
2
3 import java.io.*;
4
5 class Matrix {
6
7     int[] [] a;
8     int i, j;
9
10    void getval() throws IOException {
11
12        BufferedReader br = new BufferedReader(
13            new InputStreamReader(System.in));
14
15        System.out.print("Enter the number of rows: ");
16        i = Integer.parseInt(br.readLine());
17
18        System.out.print("Enter the number of columns: ");
19        j = Integer.parseInt(br.readLine());
20
21        a = new int[i][j];
22
23        System.out.println("Enter the values in " + i + " x " + j + " matrix:"
24    );
25
26        for (int x = 0; x < i; x++) {
27
28            for (int y = 0; y < j; y++) {
29                a[x][y] = Integer.parseInt(br.readLine());
30            }
31        }
32
33        void display() {
34
```

```

35         System.out.println("\nMatrix is:");
36
37         for (int x = 0; x < i; x++) {
38
39             for (int y = 0; y < j; y++) {
40                 System.out.print(a[x][y] + " ");
41             }
42
43             System.out.println();
44         }
45
46         System.out.println("\nTranspose of Matrix:");
47
48         for (int x = 0; x < j; x++) {
49
50             for (int y = 0; y < i; y++) {
51                 System.out.print(a[y][x] + " ");
52             }
53
54             System.out.println();
55         }
56     }
57 }
58
59 class MatrixDemo {
60
61     public static void main(String[] args) throws IOException {
62
63         Matrix m = new Matrix();
64
65         m.getval();
66         m.display();
67     }
68 }

```

Output:

```

Enter the number of rows: 3
Enter the number of columns: 3
Enter the values in 3 x 3 matrix:
1
2
3
4
5
6
7
8
9
Matrix is:

```

```
1 2 3
4 5 6
7 8 9
Transpose of Matrix:
1 4 7
2 5 8
3 6 9
```

JAVA PROGRAMMING LAB

Experiment No.: 58

Date: 13/08/2026

Aim:

WAP to display numbers from 10 to 100 with a difference of 10 using a loop.

Program:

```
1 package Program;
2
3 class Loop {
4
5     void display() {
6
7         for (int i = 10; i <= 100; i = i + 10) {
8
9             System.out.println(i);
10        }
11    }
12 }
13
14 class LoopDemo {
15
16     public static void main(String[] args) {
17
18         Loop l = new Loop();
19         l.display();
20     }
21 }
```

Output:

```
10
20
30
40
50
60
70
80
90
100
```

JAVA PROGRAMMING LAB

Experiment No.: 59

Date: 13/08/2026

Aim:

WAP to find and display the square of a given number using the Math.pow() function.

Program:

```
1 package Program;
2
3 import java.lang.Math;
4 import java.util.Scanner;
5
6 class Square {
7
8     void display() {
9
10         Scanner sc = new Scanner(System.in);
11
12         System.out.print("Enter a number: ");
13         int num = sc.nextInt();
14
15         sc.close();
16
17         int sq = (int) Math.pow(num, 2);
18         System.out.println("The square of " + num + " is: " + sq);
19     }
20 }
21
22 class SquareDemo {
23
24     public static void main(String[] args) {
25
26         Square s = new Square();
27         s.display();
28     }
29 }
```

Output:

```
Enter a number: 12
The square of 12 is: 144
```

JAVA PROGRAMMING LAB

Experiment No.: 60

Date: 13/08/2026

Aim:

WAP to find and display the sine and cosine values of a given number using Math.sin() and Math.cos() functions.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class Trigonometry {
6
7     void calculate() {
8
9         Scanner sc = new Scanner(System.in);
10
11         System.out.print("Enter angle in degrees: ");
12         double degree = sc.nextDouble();
13
14         double radian = Math.toRadians(degree);
15
16         double sine = Math.sin(radian);
17         double cosine = Math.cos(radian);
18
19         System.out.println("Sine value = " + sine);
20         System.out.println("Cosine value = " + cosine);
21
22         sc.close();
23     }
24 }
25
26 class TrigonometryDemo {
27
28     public static void main(String[] args) {
29
30         Trigonometry t = new Trigonometry();
31         t.calculate();
32     }
33 }
```

Output:

```
Enter angle in degrees: 90  
Sine value = 1.0  
Cosine value = 6.123233995736766E-17
```

JAVA PROGRAMMING LAB

Experiment No.: 61

Date: 13/08/2026

Aim:

WAP to find and display the maximum value between two given values using the Math.max() function.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class Maximum {
6     int a, b;
7
8     void accept() {
9         Scanner sc = new Scanner(System.in);
10
11         System.out.print("Enter first value: ");
12         a = sc.nextInt();
13
14         System.out.print("Enter second value: ");
15         b = sc.nextInt();
16
17         sc.close();
18     }
19
20     void display() {
21         int max = Math.max(a, b);
22
23         System.out.println("Maximum value = " + max);
24     }
25 }
26
27 class MaximumDemo {
28     public static void main(String[] args) {
29         Maximum m = new Maximum();
30
31         m.accept();
32         m.display();
33     }
34 }
```

Output:

```
Enter first value: 22  
Enter second value: 55  
Maximum value = 55
```

JAVA PROGRAMMING LAB

Experiment No.: 62

Date: 13/08/2026

Aim:

WAP to find and display the power of one number raised to another number using the Math.pow() function.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class Power {
6
7     int a, b;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter base value: ");
14
15         a = sc.nextInt();
16
17         System.out.print("Enter power value: ");
18
19         b = sc.nextInt();
20
21         sc.close();
22     }
23
24     void display() {
25
26         double result = Math.pow(a, b);
27
28         System.out.println("Power = " + result);
29
30     }
31 }
32
33
34 class PowerDemo {
35
```

```
36     public static void main(String[] args) {  
37  
38         Power p = new Power();  
39  
40         p.accept();  
41  
42         p.display();  
43  
44     }  
45  
46 }
```

Output:

```
Enter base value: 4  
Enter power value: 4  
Power = 256.0
```

JAVA PROGRAMMING LAB

Experiment No.: 63

Date: 13/08/2026

Aim:

WAP to find and display the smaller value between two given numbers using the Math.min() function.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class Minimum {
6
7     int a, b;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter first value: ");
14
15         a = sc.nextInt();
16
17         System.out.print("Enter second value: ");
18
19         b = sc.nextInt();
20
21         sc.close();
22
23     }
24
25     void display() {
26
27         int min = Math.min(a, b);
28
29         System.out.println("Smaller value = " + min);
30
31     }
32
33 }
34
35 class MinimumDemo {
```

```
36
37     public static void main(String[] args) {
38
39         Minimum m = new Minimum();
40
41         m.accept();
42
43         m.display();
44
45     }
46
47 }
```

Output:

```
Enter first value: 20
Enter second value: 50
Smaller value = 20
```

JAVA PROGRAMMING LAB

Experiment No.: 64

Date: 13/08/2026

Aim:

WAP to reverse and display a given string.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class ReverseString {
6
7     String str;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter a string: ");
14
15         str = sc.nextLine();
16
17         sc.close();
18
19     }
20
21     void display() {
22
23         String reverse = "";
24
25         for (int i = str.length() - 1; i >= 0; i--) {
26
27             reverse = reverse + str.charAt(i);
28
29         }
30
31         System.out.println("Reversed string = " + reverse);
32
33     }
34
35 }
36
```

```
37 class ReverseStringDemo {
38
39     public static void main(String[] args) {
40
41         ReverseString r = new ReverseString();
42
43         r.accept();
44
45         r.display();
46
47     }
48
49 }
```

Output:

```
Enter a string: Aditya
Reversed string = aytidA
```

JAVA PROGRAMMING LAB

Experiment No.: 65

Date: 13/08/2026

Aim:

WAP to display a given string using string methods.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class StringMethods {
6
7     String str;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter a string: ");
14
15         str = sc.nextLine();
16
17         sc.close();
18
19     }
20
21     void display() {
22
23         System.out.println("String = " + str);
24         System.out.println("Length = " + str.length());
25         System.out.println("Uppercase = " + str.toUpperCase());
26         System.out.println("Lowercase = " + str.toLowerCase());
27         System.out.println("First character = " + str.charAt(0));
28
29     }
30
31 }
32
33 class StringMethodsDemo {
34
35     public static void main(String[] args) {
36
```

```
37         StringMethods s = new StringMethods();
38
39         s.accept();
40
41         s.display();
42
43     }
44
45 }
```

Output:

```
Enter a string: Aditya
String = Aditya
Length = 6
Uppercase = ADITYA
Lowercase = aditya
First character = A
```

JAVA PROGRAMMING LAB

Experiment No.: 66

Date: 13/08/2026

Aim:

WAP to display a given string and demonstrate string manipulation.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class StringManipulation {
6
7     String str;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter a string: ");
14
15         str = sc.nextLine();
16
17         sc.close();
18
19     }
20
21     void display() {
22
23         System.out.println("Original String = " + str);
24
25         System.out.println("Length = " + str.length());
26
27         System.out.println("Uppercase = " + str.toUpperCase());
28
29         System.out.println("Lowercase = " + str.toLowerCase());
30
31         System.out.println("First character = " + str.charAt(0));
32
33         System.out.println("Substring = " + str.substring(0, str.length() / 2)
34 );
35
36         System.out.println("Replaced String = " + str.replace('a', 'A'));
```

```
36     }
37
38 }
39
40
41 class StringManipulationDemo {
42
43     public static void main(String[] args) {
44
45         StringManipulation s = new StringManipulation();
46
47         s.accept();
48
49         s.display();
50
51     }
52
53 }
```

Output:

```
Enter a string: Aditya Raj
Original String = Aditya Raj
Length = 10
Uppercase = ADITYA RAJ
Lowercase = aditya raj
First character = A
Substring = Adity
Replaced String = AdityA RAj
```

JAVA PROGRAMMING LAB

Experiment No.: 67

Date: 13/08/2026

Aim:

WAP to replace a character in a given string with another character.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class ReplaceCharacter {
6
7     String str;
8     char oldChar, newChar;
9
10    void accept() {
11
12        Scanner sc = new Scanner(System.in);
13
14        System.out.print("Enter a string: ");
15
16        str = sc.nextLine();
17
18        System.out.print("Enter character to replace: ");
19
20        oldChar = sc.next().charAt(0);
21
22        System.out.print("Enter new character: ");
23
24        newChar = sc.next().charAt(0);
25
26        sc.close();
27    }
28
29
30    void display() {
31
32        String result = str.replace(oldChar, newChar);
33
34        System.out.println("Original String = " + str);
35
36        System.out.println("Replaced String = " + result);
```

```
37  
38     }  
39  
40 }  
41  
42 class ReplaceCharacterDemo {  
43  
44     public static void main(String[] args) {  
45  
46         ReplaceCharacter r = new ReplaceCharacter();  
47  
48         r.accept();  
49  
50         r.display();  
51  
52     }  
53  
54 }
```

Output:

```
Enter a string: ranchi  
Enter character to replace: r  
Enter new character: R  
Original String = ranchi  
Replaced String = Ranchi
```

JAVA PROGRAMMING LAB

Experiment No.: 68

Date: 13/08/2026

Aim:

WAP to find and display the length of a given string.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class StringLength {
6
7     String str;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter a string: ");
14
15         str = sc.nextLine();
16
17         sc.close();
18
19     }
20
21     void display() {
22
23         int length = str.length();
24
25         System.out.println("Length of string = " + length);
26
27     }
28
29 }
30
31 class StringLengthDemo {
32
33     public static void main(String[] args) {
34
35         StringLength s = new StringLength();
36
37     }
```

```
37         s.accept();
38
39         s.display();
40
41     }
42
43 }
```

Output:

```
Enter a string: Aditya
Length of string = 6
```

JAVA PROGRAMMING LAB

Experiment No.: 69

Date: 20/08/2026

Aim:

WAP accept any two number from the keyboard and display their sum.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class SumOfTwoNumbers {
6
7     int a, b;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter first number: ");
14
15         a = sc.nextInt();
16
17         System.out.print("Enter second number: ");
18
19         b = sc.nextInt();
20
21         sc.close();
22
23     }
24
25     void display() {
26
27         int sum = a + b;
28
29         System.out.println("Sum = " + sum);
30
31     }
32
33 }
34
35 class SumOfTwoNumbersDemo {
36
```

```
37     public static void main(String[] args) {  
38  
39         SumOfTwoNumbers s = new SumOfTwoNumbers();  
40  
41         s.accept();  
42  
43         s.display();  
44  
45     }  
46  
47 }
```

Output:

```
Enter first number: 20  
Enter second number: 50  
Sum = 70
```

JAVA PROGRAMMING LAB

Experiment No.: 70

Date: 20/08/2026

Aim:

WAP to accept and display a character from the keyboard. Using scanner class.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class CharacterInput {
6
7     char ch;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter a character: ");
14
15         ch = sc.next().charAt(0);
16
17         sc.close();
18
19     }
20
21     void display() {
22
23         System.out.println("Entered character = " + ch);
24
25     }
26
27 }
28
29 class CharacterInputDemo {
30
31     public static void main(String[] args) {
32
33         CharacterInput c = new CharacterInput();
34
35         c.accept();
36
37     }
```

```
37         c.display();
38
39     }
40
41 }
```

Output:

```
Enter a character: A
The entered character is: A
```

JAVA PROGRAMMING LAB

Experiment No.: 71

Date: 20/08/2026

Aim:

WAP to accept and display a string or a name from the keyboard. Using scanner class.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class StringInput {
6
7     String name;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter your name: ");
14
15         name = sc.nextLine();
16
17         sc.close();
18
19     }
20
21     void display() {
22
23         System.out.println("Entered name = " + name);
24
25     }
26
27 }
28
29 class StringInputDemo {
30
31     public static void main(String[] args) {
32
33         StringInput s = new StringInput();
34
35         s.accept();
```

```
36
37     s.display();
38
39 }
40
41 }
```

Output:

```
Enter your name: Aditya
Your name is: Aditya
```

JAVA PROGRAMMING LAB

Experiment No.: 72

Date: 20/08/2026

Aim:

WAP to accept and display an integer from the keyboard. Using scanner class

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class IntegerInput {
6
7     int num;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter an integer: ");
14
15         num = sc.nextInt();
16
17         sc.close();
18
19     }
20
21     void display() {
22
23         System.out.println("Entered integer = " + num);
24
25     }
26
27 }
28
29 class IntegerInputDemo {
30
31     public static void main(String[] args) {
32
33         IntegerInput i = new IntegerInput();
34
35         i.accept();
36
37     }
```

```
37         i.display();
38
39     }
40
41 }
```

Output:

```
Enter an integer: 12
The entered integer is: 12
```

JAVA PROGRAMMING LAB

Experiment No.: 73

Date: 20/08/2026

Aim:

WAP to accept and display a float number from the keyboard. Using scanner class.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class FloatInput {
6
7     float num;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter a float number: ");
14
15         num = sc.nextFloat();
16
17         sc.close();
18
19     }
20
21     void display() {
22
23         System.out.println("Entered float number = " + num);
24
25     }
26
27 }
28
29 class FloatInputDemo {
30
31     public static void main(String[] args) {
32
33         FloatInput f = new FloatInput();
34
35         f.accept();
36
37     }
```

```
37         f.display();
38
39     }
40
41 }
```

Output:

```
Enter a float number: 22.34
The entered float number is: 22.34
```

JAVA PROGRAMMING LAB

Experiment No.: 74

Date: 20/08/2026

Aim:

WAP to accept and display employee details. Using scanner class.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class Employee {
6
7     int id;
8     String name;
9     float salary;
10
11     void accept() {
12
13         Scanner sc = new Scanner(System.in);
14
15         System.out.print("Enter employee ID: ");
16
17         id = sc.nextInt();
18
19         sc.nextLine();
20
21         System.out.print("Enter employee name: ");
22
23         name = sc.nextLine();
24
25         System.out.print("Enter employee salary: ");
26
27         salary = sc.nextFloat();
28
29         sc.close();
30
31     }
32
33     void display() {
34
35         System.out.println("\nEmployee Details");
36
37     }
```

```
37         System.out.println("Employee ID = " + id);
38
39         System.out.println("Employee Name = " + name);
40
41         System.out.println("Employee Salary = " + salary);
42
43     }
44
45 }
46
47 class EmployeeDemo {
48
49     public static void main(String[] args) {
50
51         Employee e = new Employee();
52
53         e.accept();
54
55         e.display();
56
57     }
58
59 }
```

Output:

```
Enter empid: 101
Enter empname: Aditya
Enter salary: 100000
Employee Details
Employee ID: 101
Employee Name: Aditya
Employee Salary: 100000.0
```

JAVA PROGRAMMING LAB

Experiment No.: 75

Date: 20/08/2026

Aim:

WAP to accept any integer from the keyboard and test if it is even or not.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class Even {
6
7     int num;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter an integer: ");
14
15         num = sc.nextInt();
16
17         sc.close();
18
19     }
20
21     void display() {
22
23         if (num % 2 == 0) {
24
25             System.out.println(num + " is an even number.");
26
27         } else {
28
29             System.out.println(num + " is not an even number.");
30
31         }
32
33     }
34
35 }
36
```

```
37 class EvenDemo {
38
39     public static void main(String[] args) {
40
41         Even e = new Even();
42
43         e.accept();
44
45         e.display();
46
47     }
48
49 }
```

Output:

```
Enter an integer: 24
24 is an even number.
```

JAVA PROGRAMMING LAB

Experiment No.: 76

Date: 20/08/2026

Aim:

WAP to accept any two float number from the keyboard and display their sum.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class FloatSum {
6
7     float a, b;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter first float number: ");
14
15         a = sc.nextFloat();
16
17         System.out.print("Enter second float number: ");
18
19         b = sc.nextFloat();
20
21         sc.close();
22
23     }
24
25     void display() {
26
27         float sum = a + b;
28
29         System.out.println("Sum = " + sum);
30
31     }
32
33 }
34
35 class FloatSumDemo {
36
```

```
37     public static void main(String[] args) {  
38  
39         FloatSum f = new FloatSum();  
40  
41         f.accept();  
42  
43         f.display();  
44  
45     }  
46  
47 }
```

Output:

```
Enter first float number: 33.45  
Enter second float number: 65.54  
Sum = 98.990005
```

JAVA PROGRAMMING LAB

Experiment No.: 77

Date: 20/08/2026

Aim:

WAP to create a 1D array and read its elements by using a loop display them one by one.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class ArrayDisplay {
6
7     int a[];
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter size of array: ");
14
15         int n = sc.nextInt();
16
17         a = new int[n];
18
19         for (int i = 0; i < n; i++) {
20
21             System.out.print("Enter element " + (i + 1) + ": ");
22
23             a[i] = sc.nextInt();
24
25         }
26
27         sc.close();
28
29     }
30
31     void display() {
32
33         System.out.println("Array elements are:");
34
35         for (int i = 0; i < a.length; i++) {
```

```
36
37         System.out.println(a[i]);
38     }
39 }
40
41 }
42
43 }
44
45 class ArrayDisplayDemo {
46
47     public static void main(String[] args) {
48
49         ArrayDisplay a = new ArrayDisplay();
50
51         a.accept();
52
53         a.display();
54
55     }
56
57 }
```

Output:

```
Enter size of array: 3
Enter element 1: 1
Enter element 2: 2
Enter element 3: 3
Array elements are:
1
2
3
```

JAVA PROGRAMMING LAB

Experiment No.: 78

Date: 20/08/2026

Aim:

WAP to accepts the marks of a students into 1D array from the keyboard and finds total marks and percentage.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class StudentMarks {
6
7     int marks[];
8     int total;
9     float percentage;
10
11     void accept() {
12
13         Scanner sc = new Scanner(System.in);
14
15         System.out.print("Enter number of subjects: ");
16
17         int n = sc.nextInt();
18
19         marks = new int[n];
20
21         for (int i = 0; i < n; i++) {
22
23             System.out.print("Enter marks of subject " + (i + 1) + ": ");
24
25             marks[i] = sc.nextInt();
26
27         }
28
29         sc.close();
30
31     }
32
33     void display() {
34
35         total = 0;
```

```
36
37     for (int i = 0; i < marks.length; i++) {
38
39         total = total + marks[i];
40
41     }
42
43     percentage = (float) total / marks.length;
44
45     System.out.println("Total Marks = " + total);
46
47     System.out.println("Percentage = " + percentage + "%");
48
49 }
50
51 }
52
53 class StudentMarksDemo {
54
55     public static void main(String[] args) {
56
57         StudentMarks s = new StudentMarks();
58
59         s.accept();
60
61         s.display();
62
63     }
64
65 }
```

Output:

```
Enter number of subjects: 4
Enter marks of subject 1: 88
Enter marks of subject 2: 67
Enter marks of subject 3: 98
Enter marks of subject 4: 59
Total Marks = 312
Percentage = 78.0%
```

JAVA PROGRAMMING LAB

Experiment No.: 79

Date: 20/08/2026

Aim:

WAP to accept any integer from the keyboard and check whether it is positive, negative, or zero.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class NumberCheck {
6
7     int num;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter an integer: ");
14
15         num = sc.nextInt();
16
17         sc.close();
18
19     }
20
21     void display() {
22
23         if (num > 0) {
24
25             System.out.println(num + " is a positive number.");
26
27         } else if (num < 0) {
28
29             System.out.println(num + " is a negative number.");
30
31         } else {
32
33             System.out.println("The number is zero.");
34
35         }
```

```
36
37     }
38
39 }
40
41 class NumberCheckDemo {
42
43     public static void main(String[] args) {
44
45         NumberCheck n = new NumberCheck();
46
47         n.accept();
48
49         n.display();
50
51     }
52
53 }
```

Output:

```
Enter an integer: 45
45 is a positive number.
```

JAVA PROGRAMMING LAB

Experiment No.: 80

Date: 20/08/2026

Aim:

WAP to accept any two integers from the keyboard and display the larger number.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class LargerNumber {
6
7     int a, b;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter first integer: ");
14
15         a = sc.nextInt();
16
17         System.out.print("Enter second integer: ");
18
19         b = sc.nextInt();
20
21         sc.close();
22
23     }
24
25     void display() {
26
27         if (a > b) {
28
29             System.out.println("Larger number = " + a);
30
31         } else if (b > a) {
32
33             System.out.println("Larger number = " + b);
34
35         } else {
36
```

```
37         System.out.println("Both numbers are equal.");
38     }
39 }
40
41 }
42
43 }
44
45 class LargerNumberDemo {
46
47     public static void main(String[] args) {
48
49         LargerNumber l = new LargerNumber();
50
51         l.accept();
52
53         l.display();
54
55     }
56
57 }
```

Output:

```
Enter first integer: 55
Enter second integer: 99
Larger number = 99
```

JAVA PROGRAMMING LAB

Experiment No.: 81

Date: 20/08/2026

Aim:

WAP to accept any integer from the keyboard and check whether it is odd or even.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class OddEven {
6
7     int num;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter an integer: ");
14
15         num = sc.nextInt();
16
17         sc.close();
18
19     }
20
21     void display() {
22
23         if (num % 2 == 0) {
24
25             System.out.println(num + " is an even number.");
26
27         } else {
28
29             System.out.println(num + " is an odd number.");
30
31         }
32
33     }
34
35 }
36
```

```
37 class OddEvenDemo {
38
39     public static void main(String[] args) {
40
41         OddEven o = new OddEven();
42
43         o.accept();
44
45         o.display();
46
47     }
48
49 }
```

Output:

```
Enter an integer: 36
36 is an even number.
```

JAVA PROGRAMMING LAB

Experiment No.: 82

Date: 20/08/2026

Aim:

WAP to accept three numbers from the keyboard and display the largest number.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class LargestNumber {
6
7     int a, b, c;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter first number: ");
14
15         a = sc.nextInt();
16
17         System.out.print("Enter second number: ");
18
19         b = sc.nextInt();
20
21         System.out.print("Enter third number: ");
22
23         c = sc.nextInt();
24
25         sc.close();
26
27     }
28
29     void display() {
30
31         if (a >= b && a >= c) {
32
33             System.out.println("Largest number = " + a);
34
35         } else if (b >= a && b >= c) {
36
```

```
37         System.out.println("Largest number = " + b);
38
39     } else {
40
41         System.out.println("Largest number = " + c);
42
43     }
44
45 }
46
47 }
48
49 class LargestNumberDemo {
50
51     public static void main(String[] args) {
52
53         LargestNumber l = new LargestNumber();
54
55         l.accept();
56
57         l.display();
58
59     }
60
61 }
```

Output:

```
Enter first number: 33
Enter second number: 45
Enter third number: 99
Largest number = 99
```

JAVA PROGRAMMING LAB

Experiment No.: 83

Date: 20/08/2026

Aim:

WAP to accept a number from the keyboard and display its multiplication table up to 10.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class MultiplicationTable {
6
7     int num;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter a number: ");
14
15         num = sc.nextInt();
16
17         sc.close();
18
19     }
20
21     void display() {
22
23         for (int i = 1; i <= 10; i++) {
24
25             System.out.println(num + " x " + i + " = " + (num * i));
26
27         }
28
29     }
30
31 }
32
33 class MultiplicationTableDemo {
34
35     public static void main(String[] args) {
```

```
36
37     MultiplicationTable m = new MultiplicationTable();
38
39     m.accept();
40
41     m.display();
42
43 }
44
45 }
```

Output:

```
Enter a number: 20
20 x 1 = 20
20 x 2 = 40
20 x 3 = 60
20 x 4 = 80
20 x 5 = 100
20 x 6 = 120
20 x 7 = 140
20 x 8 = 160
20 x 9 = 180
20 x 10 = 200
```

JAVA PROGRAMMING LAB

Experiment No.: 84

Date: 20/08/2026

Aim:

WAP to accept an integer from the keyboard and find and display the sum of its digits.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class DigitSum {
6
7     int num;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter an integer: ");
14
15         num = sc.nextInt();
16
17         sc.close();
18
19     }
20
21     void display() {
22
23         int n = num;
24         int sum = 0;
25
26         while (n != 0) {
27
28             int digit = n % 10;
29
30             sum = sum + digit;
31
32             n = n / 10;
33
34         }
35
36     }
```

```
36         System.out.println("Sum of digits = " + sum);
37     }
38 }
39
40 }
41
42 class DigitSumDemo {
43
44     public static void main(String[] args) {
45
46         DigitSum d = new DigitSum();
47
48         d.accept();
49
50         d.display();
51
52     }
53
54 }
```

Output:

```
Enter an integer: 123
Sum of digits = 6
```

JAVA PROGRAMMING LAB

Experiment No.: 85

Date: 20/08/2026

Aim:

WAP to accept an integer from the keyboard and find and display its factorial.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class Factorial {
6
7     int num;
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter an integer: ");
14
15         num = sc.nextInt();
16
17         sc.close();
18
19     }
20
21     void display() {
22
23         int fact = 1;
24
25         for (int i = 1; i <= num; i++) {
26
27             fact = fact * i;
28
29         }
30
31         System.out.println("Factorial = " + fact);
32
33     }
34
35 }
36
```

```
37 class FactorialDemo {  
38  
39     public static void main(String[] args) {  
40  
41         Factorial f = new Factorial();  
42  
43         f.accept();  
44  
45         f.display();  
46     }  
47 }  
48  
49 }
```

Output:

```
Enter an integer: 5  
Factorial = 120
```

JAVA PROGRAMMING LAB

Experiment No.: 86

Date: 20/08/2026

Aim:

WAP to create a 1D array, accept its elements from the keyboard, and find and display the largest element.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class LargestArrayElement {
6
7     int a[];
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter size of array: ");
14
15         int n = sc.nextInt();
16
17         a = new int[n];
18
19         for (int i = 0; i < n; i++) {
20
21             System.out.print("Enter element " + (i + 1) + ": ");
22
23             a[i] = sc.nextInt();
24
25         }
26
27         sc.close();
28
29     }
30
31     void display() {
32
33         int largest = a[0];
34
35         for (int i = 1; i < a.length; i++) {
```

```

36         if (a[i] > largest) {
37             largest = a[i];
38         }
39     }
40
41     System.out.println("Largest element = " + largest);
42 }
43
44 class LargestArrayElementDemo {
45     public static void main(String[] args) {
46         LargestArrayElement l = new LargestArrayElement();
47         l.accept();
48         l.display();
49     }
50 }

```

Output:

```

Enter size of array: 5
Enter element 1: 1
Enter element 2: 2
Enter element 3: 3
Enter element 4: 4
Enter element 5: 5
Largest element = 5

```

JAVA PROGRAMMING LAB

Experiment No.: 87

Date: 20/08/2026

Aim:

WAP to create a 1D array, accept its elements from the keyboard, and find and display the smallest element.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class SmallestArrayElement {
6
7     int a[];
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter size of array: ");
14
15         int n = sc.nextInt();
16
17         a = new int[n];
18
19         for (int i = 0; i < n; i++) {
20
21             System.out.print("Enter element " + (i + 1) + ": ");
22
23             a[i] = sc.nextInt();
24
25         }
26
27         sc.close();
28
29     }
30
31     void display() {
32
33         int smallest = a[0];
34
35         for (int i = 1; i < a.length; i++) {
```

```

36         if (a[i] < smallest) {
37
38             smallest = a[i];
39
40         }
41     }
42 }
43
44     System.out.println("Smallest element = " + smallest);
45
46 }
47
48 }
49
50
51 class SmallestArrayElementDemo {
52
53     public static void main(String[] args) {
54
55         SmallestArrayElement s = new SmallestArrayElement();
56
57         s.accept();
58
59         s.display();
60
61     }
62
63 }

```

Output:

```

Enter size of array: 5
Enter element 1: 5
Enter element 2: 4
Enter element 3: 3
Enter element 4: 2
Enter element 5: 1
Smallest element = 1

```

JAVA PROGRAMMING LAB

Experiment No.: 88

Date: 20/08/2026

Aim:

WAP to create a 1D array, accept its elements from the keyboard, and find and display the sum and average of all elements.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class ArraySumAverage {
6
7     int a[];
8
9     void accept() {
10
11         Scanner sc = new Scanner(System.in);
12
13         System.out.print("Enter size of array: ");
14
15         int n = sc.nextInt();
16
17         a = new int[n];
18
19         for (int i = 0; i < n; i++) {
20
21             System.out.print("Enter element " + (i + 1) + ": ");
22
23             a[i] = sc.nextInt();
24
25         }
26
27         sc.close();
28
29     }
30
31     void display() {
32
33         int sum = 0;
34
35         for (int i = 0; i < a.length; i++) {
```

```

36
37         sum = sum + a[i];
38
39     }
40
41     float average = (float) sum / a.length;
42
43     System.out.println("Sum = " + sum);
44
45     System.out.println("Average = " + average);
46
47 }
48
49 }
50
51 class ArraySumAverageDemo {
52
53     public static void main(String[] args) {
54
55         ArraySumAverage a = new ArraySumAverage();
56
57         a.accept();
58
59         a.display();
60
61     }
62
63 }

```

Output:

```

Enter size of array: 5
Enter element 1: 1
Enter element 2: 2
Enter element 3: 3
Enter element 4: 4
Enter element 5: 5
Sum = 15
Average = 3.0

```